

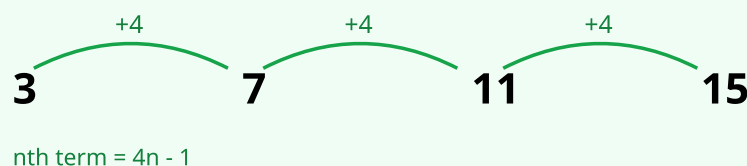
Arithmetic Sequences — nth Term

AQA · KS3 Year 7–8

Name: _____ Class: _____ Date: _____

Find and use the nth term rule (of the form $an + b$) for a linear (arithmetic) sequence. Use it to generate terms and to decide whether a given number is in the sequence.

VISUAL EXAMPLE



Arithmetic sequence: a constant difference is added each step.

COMMON MISTAKES

Writing the rule as $d + n$ instead of dn (the difference multiplies n).

Getting the sign of the constant wrong — find it from (first term - common difference).

Confusing the term number (n) with the term value.

Saying a number is in the sequence without checking that n is a positive whole number.

METHOD

- 1 Find the common difference d (subtract a term from the next term).
- 2 The rule begins with dn (the difference times n).
- 3 Find the constant b = first term - d .
- 4 Write the nth term as $dn + b$.
- 5 Check by substituting $n = 1, 2, 3$ to make sure you get the sequence back.

WORKED EXAMPLE

Find the nth term of: 3, 7, 11, 15, ...

Step 1 — Common difference: $7 - 3 = 4$, so $d = 4$.

Step 2 — Rule begins with $4n$.

Step 3 — Constant: first term - $d = 3 - 4 = -1$.

Step 4 — nth term = $4n - 1$.

Step 5 — Check $n = 1$: $4(1) - 1 = 3$; $n = 2$: $4(2) - 1 = 7$

1 Write the next two terms of each sequence.

(2 marks)

a) 2, 5, 8, 11, ...

b) 20, 17, 14, ...

HOW TO START

Look at how much the sequence goes up (or down) each step.

Add that same amount to the last term to get the next terms.

ANSWERS

(a) = _____

(b) = _____

METHOD 1 Find the common difference d | 2 Rule starts with dn | 3 Constant = first term - d | 4 n th term = dn + constant | 5 Check with $n = 1, 2, 3$

2 Find the common difference and the nth term.

(2 marks)

a) 4, 6, 8, 10, ...

b) 1, 5, 9, 13, ...

HOW TO START

The rule starts with the common difference times n .

For 4, 6, 8, 10: $d = 2$, so it starts $2n$; constant = $4 - 2 = 2$, giving $2n + 2$.

ANSWERS

(a) = _____

(b) = _____

METHOD 1 Find the common difference d | 2 Rule starts with dn | 3 Constant = first term - d | 4 nth term = $dn + \text{constant}$ | 5 Check with $n = 1, 2, 3$

3 Find the n th term of each sequence.

(3 marks)

a) 5, 8, 11, 14, ...

b) 7, 12, 17, 22, ...

c) 10, 8, 6, 4, ...

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Find the common difference d | 2 Rule starts with dn | 3 Constant = first term - d | 4 n th term = dn + constant | 5 Check with $n = 1, 2, 3$

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(3 marks)

- Find the 10th term.
- Find the 20th term.
- Which term has the value 51?

ANSWERS

(a) = _____

(b) =

(c) = _____

METHOD 1 Find the common difference d | 2 Rule starts with dn | 3 Constant = first term - d | 4 n th term = $dn + \text{constant}$ | 5 Check with $n = 1, 2, 3$

5 Decide whether each number is in the sequence. Show your working.

(3 marks)

a) Is 100 in the sequence with nth term $3n + 1$?

b) Is 50 in the sequence with nth term $4n - 2$?

c) Is 30 in the sequence with nth term $5n + 2$?

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Find the common difference d | 2 Rule starts with dn | 3 Constant = first term - d | 4 nth term = $dn + \text{constant}$ | 5 Check with $n = 1, 2, 3$

(3 marks)

- a) Find a rule for the number of seats in row n . (2 marks)
b) How many seats are in row 10? (1 mark)

(a) = _____

(b) = _____

METHOD 1 Find the common difference d | 2 Rule starts with dn | 3 Constant = first term - d | 4 n th term = dn + constant | 5 Check with $n = 1, 2, 3$

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(4 marks)

- a) Write down the first three terms. (1 mark)

- b) Find the 50th term. (1 mark)

- c) Show that 201 is NOT a term in this sequence. (2 marks)

ANSWERS

(a) =

(b) =

(c) =

METHOD 1 Find the common difference d | 2 Rule starts with dn | 3 Constant = first term - d | 4 n th term = dn + constant | 5 Check with $n = 1, 2, 3$

8 Sequence A has n th term $4n + 1$. Sequence B has n th term $3n + 7$.

(5 marks)

- a) Write the first five terms of each sequence. (2 marks)
- b) Which value appears in both of your lists? (1 mark)
- c) Solve $4n + 1 = 3n + 7$ and explain what your answer tells you about the two sequences. (2 marks)

ANSWERS

- (a) = _____
- (b) = _____
- (c) = _____

METHOD 1 Find the common difference d | 2 Rule starts with dn | 3 Constant = first term - d | 4 n th term = $dn + \text{constant}$ | 5 Check with $n = 1, 2, 3$

Self-reflection

How confident do you feel with this topic now? (tick one)

Not yet

Getting there

Confident

Really confident

Which question did you find the hardest, and why?

How do I find the constant in the n th term rule?

One thing I will practise before the next lesson:

Teacher key

Q1: a) 14, 17 b) 11, 8

Q2: a) $d = 2$, n th term $= 2n + 2$ b) $d = 4$, n th term $= 4n - 3$

Q3: a) $3n + 2$ b) $5n + 2$ c) $-2n + 12$ (or $12 - 2n$)

Q4: a) $4(10) - 1 = 39$ b) $4(20) - 1 = 79$ c) $4n - 1 = 51$ $4n = 52$ $n = 13$ (the 13th term)

Q5: a) $3n + 1 = 100$ $3n = 99$ $n = 33$, whole number, so YES b) $4n - 2 = 50$ $4n = 52$ $n = 13$, so YES c) $5n + 2 = 30$ $5n = 28$ $n = 5.6$, not whole, so NO

Q6: a) $d = 3$; constant $= 15 - 3 = 12$; rule $= 3n + 12$ b) $3(10) + 12 = 42$ seats

Q7: a) 2, 8, 14 b) $6(50) - 4 = 296$ c) $6n - 4 = 201$ $6n = 205$ $n = 34.17$, not a whole number, so 201 is not a term

Q8: a) A: 5, 9, 13, 17, 21; B: 10, 13, 16, 19, 22 b) 13 c) $4n + 1 = 3n + 7$ $n = 6$; at the 6th term both sequences give the same value ($4 \times 6 + 1 = 3 \times 6 + 7 = 25$)
